Product Requirements Document

AI-Powered Study Planner for College Students

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1. Executive Summary

The AI-Powered Study Planner is a mobile-and-web product that generates, adapts, and continuously re-optimizes personalized study schedules for college students. Unlike calendars, to-do lists, or flashcard apps, the product understands what a student needs to learn, how long it will realistically take them, and when they learn best — then builds and maintains a living study plan around that understanding.

Students today juggle 4-6 courses, fragmented deadlines across LMS platforms, and study tools that require heavy manual upkeep. When plans break — which they always do — students lack the time or energy to rebuild them, and fall back on last-minute cramming. This erodes retention, increases anxiety, and depresses academic performance.

The opportunity: use AI to close the loop between what a student must learn and how they actually spend their study time — through intelligent task decomposition, dynamic rescheduling, personalized time estimation, and learning-science-backed sequencing (spaced repetition, interleaving, active recall).

This PRD defines the product requirements for an MVP that targets undergraduate students preparing for exams and managing weekly coursework, with a phased roadmap toward full semester planning, LMS integration, and predictive readiness scoring.

2. Problem Statement

College students face a persistent challenge in effectively managing their academic workload across multiple courses, deadlines, and competing priorities. Despite access to calendars and to-do lists, most students struggle to create realistic, adaptive study plans that account for their actual learning pace, cognitive load, and the varying difficulty of different subjects.

Core problem: Students know what they need to study but lack the ability to determine when, how much, and in what sequence to study — leading to chronic procrastination, last-minute cramming, burnout, and suboptimal academic performance.

Why This Problem Matters

- 60–70% of college students report feeling overwhelmed by their workload at least once per semester.
- Poor time management is consistently cited as a top predictor of academic underperformance and dropout risk.
- Cramming, the default fallback when planning fails, is estimated to reduce long-term retention by up to 50% compared to spaced learning.
- Academic stress is a leading cause of anxiety among college students, with direct links to burnout.
- The addressable market spans 20M+ U.S. college students and 200M+ globally, within an EdTech market projected to exceed $400B by 2028.

Universities provide syllabi and deadlines but rarely teach students how to plan or study effectively, leaving a systemic gap that existing point-solution tools do not close.

3. Goals

3.1 Product Goals

- Give students a single, always-current view of everything they need to study, automatically prioritized.
- Generate realistic, personalized study plans that adapt automatically when reality diverges from plan.
- Reduce cramming by shifting study effort earlier and distributing it across time (spaced practice).
- Lower the cognitive overhead of daily planning so students spend their energy studying, not re-planning.
- Provide an honest, data-backed signal of exam readiness before it's too late to act on it.

3.2 Business Goals

- Establish a defensible position at the intersection of planning intelligence and learning science, where no incumbent currently competes.
- Drive strong weekly retention through daily-use habit formation (the planner becomes the student's home screen).
- Build a funnel toward a freemium-to-paid conversion model and, longer term, institutional licensing.
- Generate a proprietary dataset on study behavior and time-estimation accuracy that compounds product quality over time.

4. Success Metrics

Metric	Definition	Target (6 months post-launch)
Plan adherence rate	% of scheduled study sessions completed as planned	≥ 60%
Cramming reduction	Shift in study-session distribution toward >48h before deadline, vs. baseline	≥ 25% reduction in day-of/day-before sessions
Weekly active usage (WAU/MAU)	Users active in a given week over monthly active users	≥ 45%
Retention (W1 / W4)	% of new users still active 1 week / 4 weeks after signup	W1 ≥ 55%, W4 ≥ 30%
Self-reported stress change	Pre/post survey on academic-planning-related anxiety	Statistically significant decrease
Estimation accuracy	Model's predicted vs. actual time-on-task, mean absolute error	≤ 20% MAE by week 6 of use

Metric	Definition	Target (6 months post-launch)
GPA / grade correlation (exploratory)	Self-reported grade outcomes among active users vs. control cohort	Directionally positive, tracked longitudinally

5. User Personas

5.1 Primary Persona — The Overwhelmed Undergrad

Attribute	Description
Demographics	18–24 years old, full-time student at a 4-year university
Course load	4–6 courses per semester across multiple disciplines
Tech comfort	High; uses smartphone, laptop, and 5+ apps daily
Current tools	Google Calendar, Notion, Apple Reminders, handwritten planners — used inconsistently and in combination
Motivation	Wants good grades, struggles with execution, frequently feels behind
Core need	A plan that tells them what to do next, and adjusts when life gets in the way

5.2 Secondary Personas

Persona	Context & Need
Graduate Student	Balances coursework, research, and teaching duties; needs long-horizon planning across months, not just weeks.
Working Student	Part-time job or internship creates unpredictable availability; needs highly flexible, constraint-aware scheduling.
Student with ADHD / Executive Function Challenges	Struggles with task initiation, time estimation, and sustained focus; needs strong external scaffolding and low-friction nudges.
International Student	May face language barriers in dense reading assignments; benefits from pacing recommendations and buffer time.

6. User Journey

6.1 Onboarding Journey

1. Student signs up and connects (or manually enters) courses, syllabi, and existing calendar commitments.
2. Student completes a short preferences survey: peak energy hours, preferred session length, subjects that feel hardest.
3. AI parses syllabi/deadlines and generates an initial multi-week study plan, surfaced as a reviewable draft.
4. Student reviews, adjusts, and confirms the plan; onboarding completes with the first day's sessions visible.

6.2 Core Usage Loop

5. Student opens the app and sees today's prioritized study sessions (not a full overwhelming list).
6. Student starts a session; timer and task context (e.g., "solve 10 calculus problems, set 3") are shown.
7. Student completes, skips, or reschedules the session; outcome is logged.
8. AI updates the model of the student's pace and automatically rebalances the remaining plan for the week.
9. Ahead of exams, the student sees a readiness indicator and any recommended catch-up sessions.

6.3 Recovery Journey (Plan Breaks)

10. Student misses several sessions in a row (e.g., due to illness or a busy week).
11. AI detects the slippage and proactively surfaces a rebalanced plan rather than waiting to be asked.
12. Student reviews a short list of trade-offs (e.g., "compress Chapter 4 review" or "extend into the weekend") and accepts one.
13. Plan is silently rebuilt; no manual re-entry of tasks is required.

7. User Stories

7.1 Onboarding & Setup

ID	As a...	I want to...	So that...
US-01	New student	add my courses and upload or link my syllabi	the app knows what I need to study without manual data entry
US-02	New student	tell the app my energy patterns and preferred study times	my plan is built around when I actually focus best
US-03	New student	review and edit the AI's first draft plan before accepting it	I trust the plan reflects my real constraints

7.2 Daily Planning & Execution

ID	As a...	I want to...	So that...
US-04	Student	see a short, prioritized list of what to study today	I don't have to decide what matters most on my own
US-05	Student	start a focused study session with a built-in timer	I can execute without switching to a separate focus app
US-06	Student	mark a session complete, partially done, or skipped	the plan reflects reality, not just intentions
US-07	Student	have my plan automatically adjust when I miss sessions	I don't have to manually rebuild my schedule from scratch

7.3 Learning Optimization

ID	As a...	I want to...	So that...
US-08	Student	have review sessions automatically spaced out over time	I retain material instead of forgetting it right after cramming
US-09	Student	see a readiness score before an exam	I know whether I need to study more or I'm actually prepared
US-10	Student	get prompted with active-recall questions instead of just re-reading notes	my study time is more effective

7.4 Notifications & Motivation

ID	As a...	I want to...	So that...
US-11	Student	get a gentle nudge when I'm falling behind on a course	I can course-correct before it becomes a crisis
US-12	Student with ADHD	receive reminders that break tasks into small, concrete first steps	starting doesn't feel as overwhelming

8. Functional Requirements

ID	Requirement	Priority
FR-01	System shall allow students to manually add courses, assignments, and exam dates.	Must

ID	Requirement	Priority
FR-02	System shall parse uploaded syllabi (PDF/DOCX) to auto-extract deadlines and topics.	Should
FR-03	System shall support two-way sync with Google Calendar and Apple Calendar for busy-time awareness.	Must
FR-04	System shall decompose large tasks (e.g., "study for midterm") into scheduled, time-boxed sessions.	Must
FR-05	System shall generate an initial multi-week study plan from courses, deadlines, and stated preferences.	Must
FR-06	System shall let students mark sessions complete, partial, skipped, or rescheduled.	Must
FR-07	System shall automatically rebalance the remaining plan when sessions are missed or completed early.	Must
FR-08	System shall learn per-user, per-task-type time estimates from historical completion data.	Should
FR-09	System shall schedule higher-difficulty material during the student's self-reported peak energy hours.	Should
FR-10	System shall apply spaced-repetition scheduling to review sessions for memorization-heavy content.	Should
FR-11	System shall compute and display a pre-exam readiness score from completed sessions and self-assessments.	Should
FR-12	System shall send proactive notifications when a student falls behind a defined threshold.	Must
FR-13	System shall support natural-language input (e.g., "chem exam on the 25th, haven't started") to generate an ad hoc plan.	Could
FR-14	System shall let students set hard constraints (job hours, commitments) that the planner must schedule around.	Must
FR-15	System shall provide a single daily view showing only that day's prioritized sessions.	Must

9. Non-functional Requirements

Category	Requirement
Performance	Plan generation and rebalancing shall complete in under 3 seconds for a typical course load (4–6 courses).
Availability	Core scheduling and session-tracking features shall maintain 99.5% uptime.
Scalability	Architecture shall support scaling from launch cohort to 500K+ users without redesign.
Data privacy	Student academic data (grades, self-assessments, course content) shall be encrypted at rest and in transit; compliant with FERPA (U.S.) and GDPR where applicable.
Accessibility	Mobile and web apps shall meet WCAG 2.1 AA standards to support students with disabilities.
Cross-platform	Product shall be available on iOS, Android, and responsive web from MVP launch.
Offline support	Today's session list and timer shall remain usable without network connectivity; sync resumes when online.
Reliability of AI output	Generated plans shall always be reviewable and editable by the student before being finalized — AI output is never silently authoritative.

10. MVP Features

The MVP is scoped narrow and deep: exam and weekly-workload planning for a single student, across manually or syllabus-entered courses, with adaptive rescheduling as the core differentiator.

- Manual course, assignment, and exam entry
- Syllabus upload with AI-assisted deadline/topic extraction
- Calendar sync (Google Calendar, Apple Calendar) for busy-time awareness
- AI-generated initial study plan from courses, deadlines, and stated preferences
- Daily prioritized session view with built-in focus timer
- Session completion tracking (complete / partial / skipped / rescheduled)
- Automatic plan rebalancing when sessions are missed or completed early
- Basic spaced-repetition scheduling for review sessions
- Falling-behind notifications and nudges
- Simple pre-exam readiness indicator based on completed sessions

11. Features Out of Scope (MVP)

- Direct LMS integration (Canvas, Blackboard, Moodle) — planned for a later phase
- Built-in flashcard/spaced-repetition content authoring (initial version schedules review time, not content)
- Collaborative or group study planning
- Predictive GPA modeling
- Institutional/admin dashboards for advisors or professors
- Native integration with focus/Pomodoro apps beyond the built-in timer
- Multi-language syllabus parsing beyond English

12. Future Enhancements

- Direct LMS integration for automatic deadline and grade ingestion
- Predictive readiness scoring calibrated against actual exam outcomes
- Built-in active-recall question generation from uploaded notes/slides
- Long-horizon planning for graduate students (thesis milestones, research + teaching load)
- Institutional licensing with advisor-facing early-warning dashboards (opt-in, privacy-preserving)
- Social/accountability features (opt-in study groups, shared deadlines)
- Wearable integration for energy/sleep-aware scheduling

13. Risks

Risk	Impact	Mitigation
AI-generated plans are inaccurate or feel untrustworthy, causing early churn	High	Always show plans as editable drafts; rapidly incorporate completion feedback to improve estimates; set conservative initial estimates
Low-friction data entry is hard to achieve (manual course setup is a drop-off point)	High	Prioritize syllabus parsing and calendar import; keep manual entry as a fast fallback, not the default path

Risk	Impact	Mitigation
Notification fatigue leads students to disable nudges entirely	Medium	Cap nudge frequency; make thresholds and tone user-configurable; test intervention timing carefully
Privacy concerns around academic and behavioral data limit adoption or trigger institutional pushback	Medium	FERPA/GDPR-compliant data handling from day one; transparent data-use policy; no data sale to third parties
Competing tools (Notion, Motion, Reclaim.ai) add academic-specific features	Medium	Move quickly on the learning-science differentiation (spaced repetition, readiness scoring) that professional scheduling tools are unlikely to prioritize
Time-estimation model performs poorly for atypical course types (labs, studio, project-based)	Medium	Launch with strong support for reading/problem-set-style courses; expand task-type coverage iteratively

14. Assumptions

- Students are willing to grant calendar and syllabus access in exchange for a meaningfully better planning experience.
- A student's historical session-completion data is sufficient signal to meaningfully personalize time estimates within several weeks of use.
- Students will trust and act on AI-generated plans if the plans remain visible, editable, and explainable.
- Course syllabi are available in a parseable digital format (PDF/DOCX) for the majority of target users.

- Mobile-first daily engagement is achievable without requiring desktop use for core planning tasks.
- Freemium access to core planning features is necessary to reach adoption scale before monetization.

15. Release Plan

Phase	Timeline	Scope
Phase 0 — Discovery & Design	Weeks 1–4	User research validation, design of onboarding and core planning UX, technical architecture for plan-generation engine
Phase 1 — Alpha	Weeks 5–10	Internal + closed alpha with a single university cohort; manual course entry, basic plan generation, no rebalancing yet
Phase 2 — Beta (MVP)	Weeks 11–18	Full MVP feature set: syllabus parsing, calendar sync, adaptive rebalancing, spaced repetition, readiness indicator; expanded multi-campus beta
Phase 3 — Public Launch	Weeks 19–24	General availability on iOS, Android, and web; freemium model live; instrumentation for all success metrics
Phase 4 — Post-Launch Iteration	Months 6–9	LMS integration pilot, predictive readiness calibration, expanded persona support (grad students, working students)

End of document.